
EE 582: Nonlinear Analysis and Control of Power Electronic Converters

Course Code	EE 582
Course Name	Nonlinear Analysis and Control of Power Electronic Converters
L-T-P-C	3-0-0-3
Prerequisites	EE301, EE309
Intended for	UG and PG
Distribution	Elective
Approval	recommended by the 36th BoA (Sep 2020)

Course Contents

- **Introduction**
Introduction to nonlinear models, nonlinear phenomena, common nonlinearities, qualitative behavior of linear systems, phase portraits, limit cycle oscillation, Jacobi linearization, direct method and indirect method of Lyapunov [6 Lectures]
- **Sources of Nonlinearities in Power Electronic Converters**
Nonlinearity due to switch and reactive components, nonlinearity introduced by the closed loop control, nonlinear phenomena observed in power electronic converters. [3 Lectures]
- **Control techniques for power electronic converters**
Voltage mode control, current mode control, PWM of type 1 and type 2, hysteresis control, sliding mode control, digital control, time optimal control. [7 Lectures]
- **Dynamic Modeling Techniques**
Average modelling of PWM converters, limitations of average model, Discrete-time 1-D and 2- D models. [4 Lectures]
- **Stability Analysis**
Basics of bifurcation theory and chaos, bifurcation of smooth and piecewise-smooth maps, nonstandard bifurcation in discontinuous maps, techniques of experimental investigations. [7 Lectures]
- **Describing Function Analysis of Power Electronic Converters**
Basics of describing function, describing function of different nonlinearities, stability analysis of limit cycle using Nyquist criteria. [7 Lectures]
- **Simulation Study** Matlab based project work on i) Analysis of instability introduced by the different non-linear components, ii) Nonlinear controller design to improve transient performance and stability boundary. [7 Lectures]

Text Books

1. S. Banerjee and G. C. Verghese, (Editors), **Nonlinear Phenomena in Power Electronics: Attractors, Bifurcations, Chaos, and Nonlinear Control**, IEEE Press, 2001.

References

1. H. K. Khalil, **Nonlinear Control**, Pearson Education, 2015
2. C. K. Tse, **Complex Behavior of Switching Power Converters**, CRC, 2003.
3. N. Mohan, T. M. Undeland, and W. P. Robbins, **Power Electronics Converters, Applications, and Design**, 3rd edition, Wiley India, 2008.
4. R. W. Erickson and D. Maksimovic, **Fundamentals of Power Electronics**, 2nd edition, Dordrecht, Kluwer, 2001.
5. L. Corradini, R. Zane, D. Maksimovic, P. Mattavelli, **Digital Control of High-Frequency Switched-Mode Power Converters**, John Wiley & Sons, 2015